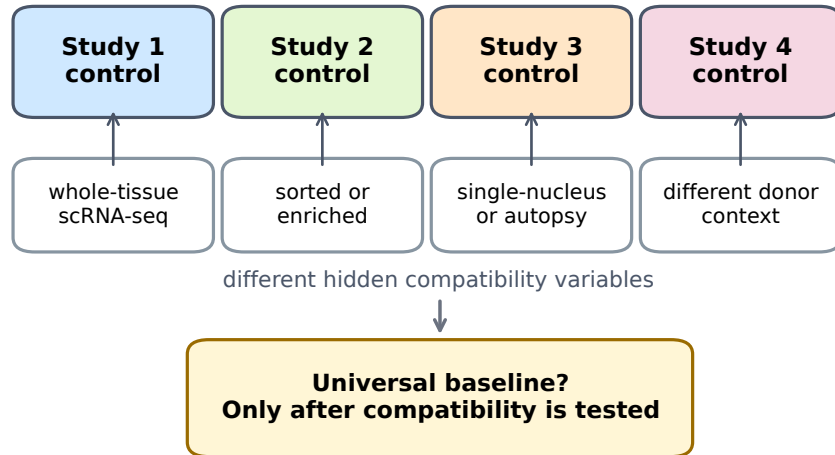


From universal-baseline illusion to compatibility-first reuse

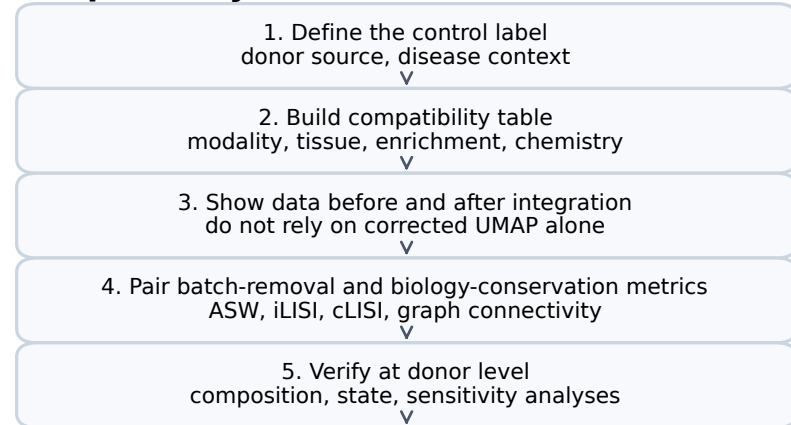
A

A shared label can hide incompatible histories



B

Compatibility-first workflow



C

Metric logic: what each diagnostic asks

Metric	Reader question	Why it matters
ASW	Are labels separated or overlapping?	Visible study or cell-type separation
iLISI	How diverse are study labels locally?	Batch mixing; higher is more mixed
cLISI	How mixed are biological labels locally?	Possible over-mixing of biological labels
Graph connectivity	Do same-label cells stay connected?	Fragmentation of cell states

D

Buyer-beware checks before pooling public controls

- [] Report unintegrated and integrated views.
- [] Separate composition-eligible datasets from sensitivity-only datasets.
- [] Treat donor or sample, not cell, as the independent unit.
- [] Use protocol-restricted and leave-one-study-out sensitivity analyses.
- [] Disclose unresolved metadata gaps rather than hiding them by integration.

Core principle:
integration aligns representations;
it does not certify study compatibility.